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Ecological wetland paradigm drives water source improvement in the stream network of Yangtze River Delta

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ABSTRACT

Jiaxing created a precedent using bypass riparian marshes to purify micro-polluted water sources in China. Pond-wetland complex with constructed root channel technology becomes a paradigm which can be analogized as “human-body wetland model” based on bionics or biomimetics. Heterogeneous plant-bed/ditch system with highly active land/water ecotone interfaces, especially meandering boundaries, breeds many biochemical reactions “living areas”. Optimization of hydraulic regulation promotes redox environment alternations and wetland treatment efficiency. Here we reported a series of upgrades and performances in Guanjinggang wetland after the Shijiuyang prototype. Morphological reform of plant-bed/ditch system played a vital role. Spatially root channel zone was main force of wetland purification, and temporally the treatment effect was higher in low-temperature seasons indicating non-temperature dependent mechanisms worked. Water pollution comprehensive index improved steadily from IV to III, and comprehensive pollution load was reduced by ca. 40%–60%. Comprehensive evaluation function value further showed the gradients purification effect of the upgraded wetland. Ecological wetlands ameliorated source water quality, and reduced drinking water treatment reagents, thereby bringing about economic benefits. Through wetlands operation, people can see how the micro-polluted surface water becomes clear and clean, so promoting a significant social benefit. As a viable component of urban green space, wetlands could beautify regional eco-environment, freshen the

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air, increase urban ecological taste, and enhance the eco-environmental protection publicity. Thus, the multifunctional service values and indirect benefits are substantial. Jiaying ecological wetlands provide a typical paradigm for water pollution remediation in developing countries and plays a leading role in technology engineering radiation effect.

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Introduction

Water plays an important role in underpinning equitable, stable, and productive societies and ecosystems (Gain et al., 2016). Provision of safe drinking water and the effective removal of bodily waste are vital for human health and well-being (Watson, 2006). Freshwater resources, which account for only a small part of the earth's surface water, are nonetheless the most precious welfare of human society (Hossain, 2015). However, with the social and economic development, our freshwater resources are constantly threatened by pollution or some emerging water environmental problems (Schwarzenbach et al., 2010; Murray et al., 2010; Fu et al., 2012; Liu et al., 2012). Thus, the purification of water and water security is challenging (Schwarzenbach et al., 2010). In China, most of the drinking water source comes from reservoirs, but some cities don't have suitable reservoirs, so they have to use river as water source (Lu et al., 2008; Fu et al., 2012). At present, groundwater and river water in China are confronted with challenging pollution problems (Qu and Fan, 2010; Han et al., 2016). In this case, Jiaying City created a precedent.

The improvement of drinking water quality is the premise and foundation of drinking water security (Qu et al., 2007). The quality of drinking-water may be controlled through a combination of protection of water sources, control of treatment processes and management of the distribution and handling of the water (World Health Organization, 2008). Ecological restoration as an important means of water source protection, the front end of drinking water security multi-barrier approach, has been paid increasing attention to during recent years (Qu and Fan, 2010; Walter et al., 2015; Wang et al., 2015b). It is of great significance to construct a multi-echelon ecological water source area with the synergetic coupling of "microorganism, plant, and ecology" (Wang et al., 2016b). The efficiency of nitrogen removal in traditional wastewater treatment wetlands is mostly poor, and at low temperature, nitrification can be insufficient to prevent a net increase in ammonia concentration due to ammonification (Akratos and Tsihrantzis, 2007; Lee et al., 2009). The key engineering technical problem is how to construct sustainable ecological wetland project to improve water quality of water source especially for ammonia nitrogen removal under low temperature. In urban and highly populated area, there are at least three major concerns about the ecological wetland solution to this problem, (1) Water flux of urban water source intake is large, (2) Urban land is worth an inch of land and the resources are limited, and (3) It is necessary to establish a Sustainable Wetland System with high reactivity and high efficiency per unit area / volume. Based on the long-term experimental researches in watershed level of finding "ecotone structure" in Baiyangdian Lake (Wang et al.,

2002; Wang and Yin, 2008) and "multipond system" in Chaohu Lake (Yin and Shan, 2001), we put forward an innovative pond-wetland ecotone complex which has been successfully applied in the plain stream network source water treatment (Wang et al., 2016b).

In this paper, we use "ecological wetland" to distinguish from artificial wetland and constructed wetland in a general sense although sometimes they are confused with each other in the broad and narrow sense (Zhang et al., 2015; Wang et al., 2017; Yang et al., 2020; Qiu et al., 2021), which is mainly based on the following considerations. Ecological wetland refers to the wetland that imitates the natural wetland in nature, especially the semi-natural wetland, such as pond (Yin and Shan, 2001), reed garden (Yin and Lan, 1995), ditch (Wang et al., 2002), landscape water body, subsided land and so on. It has most of the ecological functions of wetland and has strong hydrological regulation and filtration function. Artificial wetland refers to the wetland with strong economic purpose, such as paddy field, shallow water area of reservoir, high-yield fish pond and so on. It has a certain ecological function of wetland, which can carry out hydrological regulation and storage. And constructed wetland refers to the engineering constructed wetland, which is generally used for sewage and sludge treatment (Zhang et al., 2015; Wang et al., 2017; Qiu et al., 2021), and usually has only a single function set by human beings. The constructed "ecological wetland" not only has the core theme function of water purification, but also has multi-objective benefits or multi-win results, such as biodiversity improvement, air freshening, flood control safety, landscaping, urban wetland park, surrounding land value-added, and improvement of people's quality of life (Yin et al., 2010). Ecological wetland can achieve a series of goals such as imitating the nature, imitating ecology, sustainable ecosystem, long-term stable running, low cost, as well as low operation and maintenance.

Shijiuyang ecological wetland (110 hm², 250,000 m³/day, running since July 2008) in northern Jiaying was the first large-scale drinking water source treatment wetland in China, and has worked as a successful model for treatment of micro-polluted raw source water (Yin et al., 2010; Wang et al., 2016b). Shijiuyang ecological wetland has achieved significant environmental and ecological benefits. After the pond / root-channel wetland purification, the concentrations of source water pollutants decreased, and the major water quality indexes including ammonia nitrogen, total phosphorus, iron, manganese, petroleum, and fecal coliforms were improved significantly (Wang et al., 2016b; Su et al., 2018). Shijiuyang ecological wetland has been put into operation for about twelve years, and the treatment performance is stable without apparent media / substrate saturation or clogging. The odor and taste index of raw water in the drinking water plant

was readily improved from grade II to grade I through ecological wetland treatment, and the overall taste of drinking water was ameliorated (Yin et al., 2010). The improvement of ecological environment has been praised by leaders and the masses. Shijiuyang ecological wetland has achieved international recognition and received international / domestic awards. The Shijiuyang ecological wetland has been awarded or approved Provincial Wetland Park by the Forestry Department of Zhejiang Province in 2009, China Living Environment Model Award by the Ministry of Housing and Urban-Rural Development of the People's Republic of China (MOHURD) in 2011, the Dubai International Award for Best Practices to Improve the Living Environment in 2012 by the Dubai Municipality, Dubai-United Arab Emirates and the United Nations Human Settlements Programme (UN-HABITAT), as well as National Urban Wetland Park by MOHURD in 2013.

Under the promotion of Shijiuyang with great benefits, Jiaying further constructed an updated second such type of larger wetland—Guanjinggang ecological wetland (147 hm², 300,000 m³/day, running since October 2013) in the southern part to pretreat source water (Wang et al., 2015a, 2016a). More people will benefit from the achievements of ecological water source remediation. Here we reported as classic paradigms firstly the general functioning principles in both wetlands, secondly a series of updated technologies used in Guanjinggang and the accomplished corresponding water treatment effects, and thirdly the analogized “human-body wetland model” as well as the core mechanism and extension of water source ecological wetland.

1. Site background

The Yangtze River Delta is a densely populated and developed economic zone in China, with dense water network and rich water resources. However, in recent decades, with the rapid development of economy and the accompanying deterioration of water environment, the phenomenon of quality-oriented water shortage has generally appeared in this region, which seriously restricts the sustainable development of society and economy (Li et al., 2009; Bergman et al., 2019). Jiaying is a central city of the Yangtze River Delta city cluster and the Shanghai metropolis circle, as well as the sub-center of the Hangzhou metropolis circle. Regional socio-economic development and ecological environment governance of Jiaying are typical and representative in the Yangtze River Delta. Due to the combined pollution of upstream foreign water and its own, the water shortage in the river network area of Jiaying City is complicated and of universal significance (Yin et al., 2010).

Taking Jiaying as an example, the main problems of river network water sources are summarized as follows (Yin et al., 2010): (1) There are many (mini-) sources of pollutants and heavy pollution load in water sources. Historically, the raw water for drinking water plant has been ever directly taken from the river courses in this area, and most areas have no alternative water sources. (2) The hydrodynamic conditions in river network area are changeable. The area is low-lying, with criss-crossed water networks and numerous gates and dams. Due to the influence of tide, sluice control and rainfall, the flow direction and discharge of many rivers show complex changes

and cross pollution is serious. (3) The river network area receives the upstream guest water, and the pollutant accumulation is more in the sink, and the water purification capacity is weak. Due to the influence of human activities and anthropogenic disturbances, the natural wetlands have been almost completely lost or destroyed. In addition, the river bank has been hardened seriously, which weakens the self-purification capability of water body, and the accumulation of pollutants in the receiving water is increasingly serious.

The water quality of Jiaying area is generally from grade IV to grade V all the year round as per the environmental quality standards for surface water in China (GB 3838–2002). Most of the time, the permanganate index and ammonia nitrogen of raw water are 5–8 mg/L and 0.5–4.0 mg/L, respectively (Wang et al., 2016b). It is a typical river network with high ammonia nitrogen and high organic matter pollution (Shao, 2010; Wang et al., 2016b; Zhang et al., 2017). Therefore, in order to solve the problem of drinking water security in Jiaying, it is necessary to effectively combine the water source restoration with the advanced treatment process application in waterworks (Zhang et al., 2017). The key of water source restoration is to remove some organic matter and ammonia nitrogen, stabilize the water quality, and lay the foundation for the optimization of subsequent water plant process.

In order to solve the water quality problem of drinking water source in Jiaying City, the municipal government actively took various measures to protect and restore the water source ecosystem (Yin et al., 2010). In addition to controlling the pollution sources, it also decided to carry out the systematic construction of water source treatment project combined with the urban green space construction (Yin et al., 2010; Wang et al., 2016b). Xinchengtang water source treatment project is the first of its kind in Jiaying City. It is in the northwest corner of the urban area, with a total planning area of 2.59 km². The project includes the surrounding rural sewage treatment, dredging of polluted river sediment, construction of buffer zone in water source area and so on. The ecological wetland project area is about 110 hm² therein. The Xinchengtang ecological wetland project is technically supported by the Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences. Jiaying Water Conservancy Bureau led to complete the preliminary technical justification of the project. The engineering design was jointly completed by Water Resources & Hydroelectric Prospecting & Design Research Institute of Jiaying City and Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences. Jiaying Water Conservancy Investment Co., Ltd. is responsible for the construction, operation, maintenance, and management of the project.

Jiaying is in the southern edge of the northern subtropical zone, belonging to the East Asian monsoon region. Winter and summer monsoon alternate, with four distinct seasons, moderate temperature, abundant rain, and sunshine. It has the characteristics of wet spring, hot summer, dry autumn, and cold winter. Due to its location in the middle latitude, the hot, humid, and rainy weather in summer is much shorter than the dry and cold weather in winter (Weather, 2020). The annual average temperature is 15.9 °C. The annual average sunshine is 2017.0 hr. The average annual precipitation is 1168.6 mm. The average annual evaporation is 1327 mm (Compilation Com-

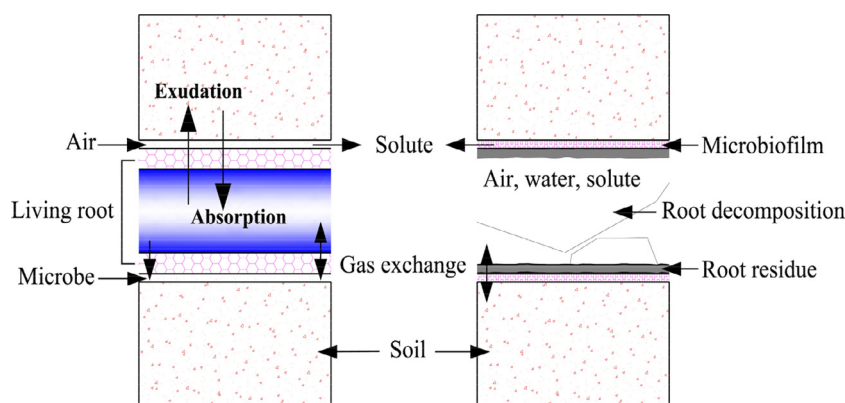


Fig. 1 – Schematic diagram of wetland root channel technology.

mittee of Jiaxing City Records, 1997). The annual evaporation is greater than the precipitation, and the climate of the city is semi-humid. Jiaxing is dominated by plain terrain.

2. Technical principle and processes

2.1. Technical principle

Water source biological, ecological restoration and water quality improvement technology mainly takes advantage of the characteristics of frequent alternation of anaerobic, anoxic, and aerobic environment and high biological activity in the land/water ecotone (Wang et al., 2002; Wang and Yin, 2008). Through the construction of large-scale land/water ecotone wetland (Yin and Lan, 1995), the construction of rich artificial root channels (identically, root passage, root entry, root holes, root pores, root pipes, etc.) in the wetland (Wang et al., 2012a) and periodic regulation of water level in operation (Wang and Yin, 2008), the water purification function of the land/water ecotone is significantly strengthened (Wang et al., 2012c). In the early stage of wetland construction, a “multilevel interface system” composed of root system, soil microorganism, water, and air as well as soil matrix was constructed (Wang et al., 2012a). The composite distribution characteristics of multi-level cross pipe holes have obvious diversion and enrichment effects on the spatial transport and migration of pollutants, and affect the flow process of various substances and energy at the soil sub-interface (Wang and Yin, 2000). The pollutants preferentially flow and migrate at the soil root channel micro-interface, and achieve the purpose of transformation and degradation and final removal in the physical, chemical, and microbial processes of the soil system (Fig. 1).

2.2. Key technology

The core of the technology lies in the innovative proposal of constructing ecological root channels (constructed root channel technology, CRCT) of constructed wetland to strengthen the water flow and material transport function of wetland (Wang et al., 2012a). CRCT simulates the natural *Phragmites australis* (common reed) root channel system in wetland, and takes the plant straw planted artificially as the filling /

medium of wetland, which effectively changes the macropore structure of wetland soil subsurface layer, and opens a new way for the construction and application of constructed wetland. The artificial root channels were used to strengthen the material transport, and the plant-bed/ditch system was constructed to increase the interface of the land/water ecotone (Wang et al., 2012a, 2012c, 2016b). Within plant bed, there are a pre-defined number of artificial root channels and a growing number of natural root channels with the succession of wetland (as CRCT illustrated in Fig. 2). The oxidation–reduction alternation process was accelerated by hydraulic regulation (Wang and Yin, 2008), and the effective interception and degradation of pollutants in the wetland system was finally realized (Fig. 2).

2.3. Technical process

Process flow: through the idea of “series and parallel connection”, the pretreatment area, the wetland root channel ecological purification area and the deep purification area form an organic series, and the large ditch, small ditch and plant bed form an organic parallel (Fig. 3).

Hydraulic regulation: reasonable design and optimal operation including hydraulic gradient and vertical design, use of semi-gate and gate to control water transfer flux, and flexible application of various hydraulic operation modes.

In order to solve the problem of water pollution in Taihu Lake Basin, an ecological constructed wetland including pretreatment area, wetland root channel ecological purification area and deep purification area was constructed (Yin et al., 2010; Wang et al., 2016b). Key purification treatment units are set up in the ecological wetland for reasonable collocation, and through the optimization of operation mode, the classification purification of various types of pollutants is realized, and the water quality of water source is significantly improved (Wang et al., 2016b). The technology can be applied to the ecological restoration of water resources in southern China (Yin et al., 2010).

2.4. Technical parameter

The conceptualized pond-wetland multistage purification system chain (Fig. 4) is composed of three principal func-

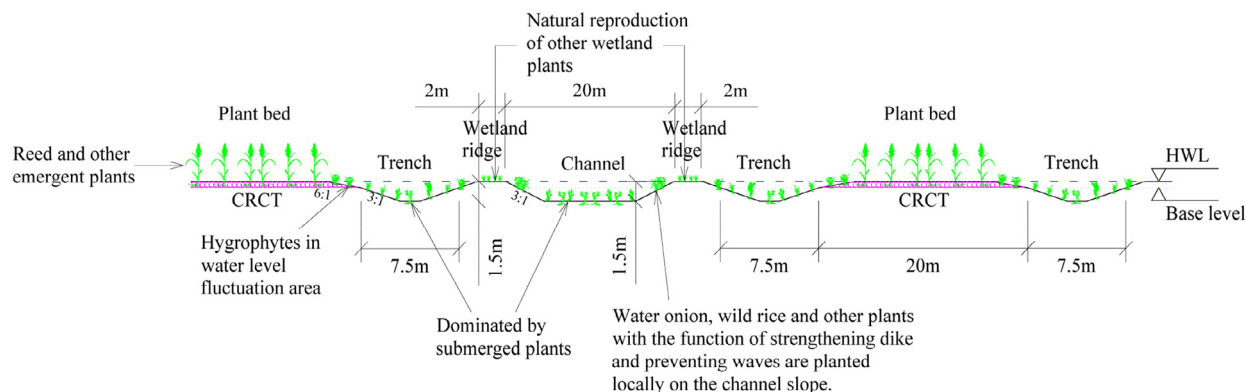


Fig. 2 – Profile of plant-bed, trench, wetland ridge, and channel continuum.

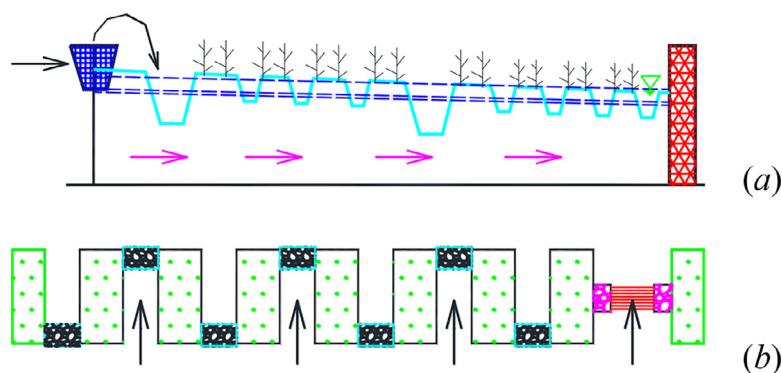


Fig. 3 – Schematic diagram of the combination of series and parallel connection of multistage wetland. (a) Series connection; (b) Parallel connection.

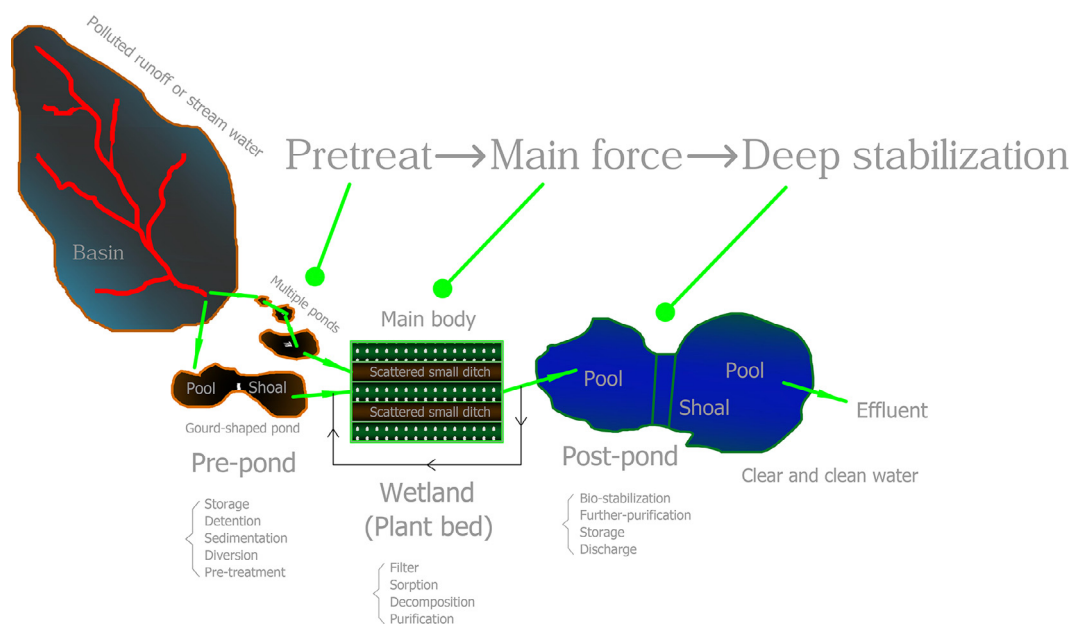


Fig. 4 – Schematic diagram of pond-wetland multistage purification system chain.

tioning zones, i.e., pretreatment area (pre-pond), wetland root channel ecological purification area (plant-bed/ditch wetland system), and deep purification area (post-pond) (Wei et al., 2018). Pretreatment area functions as the first barrier through storage, detention, sedimentation, diversion, and enhanced pretreatment. Wetland root channel ecological purification area acts as the main force of the whole wetland, mainly through filter, sorption (absorption, adsorption), decomposition, and complex purification. Deep purification area acts as the last barrier, and still serves important functions through bio-stabilization, further purification (polishing), huge storage and discharging clear and clean water into the downstream drinking water plant or receiving waters. The basic proportion of pretreatment area, wetland root channel ecological purification area, and deep purification area is approximately 3:4:3, which is a relatively ideal area proportion as far as water source ecological wetland is concerned according to the long-term project design, operation practice and actual performance. The overall hydraulic loading rate is generally not higher than $0.3\text{--}0.4\text{ m}^3/(\text{m}^2\cdot\text{day})$, and the overall hydraulic retention time is generally not lower than 3–5 days. The stored raw water within the wetland can satisfy a designed emergency buffer time of about 2–4 days in case some extreme events occur in water sources. In order to enhance the complex removal effect of various pollutants, the surface flow (through ditch, channel, pond, and pool) combined with subsurface flow (through plant bed) was adopted in the ecological wetland (Wang et al., 2016b). The total capacity of pretreatment area and deep purification area is ideally not less than 4–5 times of daily wetland treatment scale (Yin et al., 2010). In the process of designing water source ecological wetland, the emergency mechanism of dual or multi water source switching is considered. For example, the inflow of Shijiuyang ecological wetland is mainly from Xinchengtang Stream and occasionally from Beijiaohe Stream (Fig. 5 upper), and that of Guanjinggang ecological wetland is mainly from Haiyantang Stream and sometimes from Nanjiaohe Stream (Fig. 5 lower). Since the wetland is a drinking water source ecological wetland, the large capacity demand of pretreatment area and deep purification area is to ensure that the drinking water plant can still meet certain water intake under special extreme conditions. Even when all external water sources cannot be used, the effective water quantity stored in the wetland can still meet the water intake of the drinking water plant for a certain period (for instance, 2–3 days), to provide a buffer and emergency for people to cope with the extreme events.

In addition, the process of effluent re-circulating can be realized in the wetland root channel ecological purification area (Fig. 4), which can provide biodegradable organic matter from the plant-bed root exudate or litter residue for microorganisms (Wang et al., 2013b; Wei et al., 2018). On the one hand, the re-circulating process can provide organic carbon source for microbial growth and enhance microbial activity through material cycling; on the other hand, it can further remove organic pollutants especially the refractory ones. In the low temperature period during winter, the wetland treatment efficiency can be promoted by optimizing the internal management mode of the wetland. In order to prolong the effective contact time between the water body and the root channel area and improve the pollutant removal rate, the

method of one-time water inflow to high water level, outflow re-circulating through the root channel area and continual re-circulating for many days can be adopted correspondingly.

3. Guanjinggang after Shijiuyang ecological wetland

The larger-scale water source ecological wetland treatment project (Guanjinggang ecological wetland, N $30^{\circ}41'40.49''$ – $30^{\circ}42'44.65''$, E $120^{\circ}45'20.12''$ – $120^{\circ}46'45.57''$) serving Guanjinggang drinking water plant in the southern suburb of Jiaying City was designed and built by referring to the successful paradigm of water source ecological wetland (Shijiuyang ecological wetland, N $30^{\circ}46'11.98''$ – $30^{\circ}47'02.56''$, E $120^{\circ}41'50.64''$ – $120^{\circ}42'41.20''$) serving Shijiuyang drinking water plant in the northern suburb of Jiaying (Wang et al., 2015a), and Guanjinggang ecological wetland was further improved and upgraded after Shijiuyang ecological wetland (Fig. 5). The Guanjinggang ecological wetland project is in the planning wedge-shaped green space in the south of Jiaying, the northeast corner of Guanjinggang drinking water plant. The whole wetland block is bounded by Haiyantang Stream in the West, Fanggong Road in the East, Zhongnan Road in the North and the east section of the planned Nanjiaohe Stream in the South. The project covers an area of 147 hm^2 , with a daily treatment capacity of $200,000\text{--}300,000\text{ m}^3/\text{day}$ recently and $450,000\text{ m}^3/\text{day}$ in the long term.

The project was jointly designed by the Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences and Water Resources & Hydroelectric Prospecting & Design Research Institute of Jiaying City. Jiaying Water Conservancy Investment Co., Ltd. is responsible for the construction, operation, and maintenance of the project. It adopted several authorized invention patents technologies of the Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences including artificial root channel, ecological wetland, hydraulic regulation, and submerged groin (i.e., spur dike), etc. issued by the National Intellectual Property Administration, PRC.

The whole project can be divided into four sub-blocks (Fig. 5 lower). They are (I) pretreatment area, (II) wetland root channel ecological purification area, (III) deep purification area, and (IV) water diversion area. The overall process flow is as follows, Haiyantang / Nanjiaohe Stream source water → pump station lifting → (I) pretreatment area (multiple levels) → (II) wetland root channel ecological purification area (multiple levels) → connecting river section → pipe jacking project (through Haiyantang Stream underneath) → (III) deep purification area → (IV) water diversion area → water intake of waterworks. The project started construction in April 2011, began to store water and conserve aquatic plants in July 2013, and was put into trial operation in October 2013.

A series of research and demonstration projects had been carried out recently around the enhanced removal of ammonia nitrogen and organic matter from source water, as well as the enhanced removal technology of ammonia nitrogen during winter in wetland (Wang et al., 2015a, 2016a; Zhang et al., 2017; Wang et al., 2018; Su et al., 2018, 2019). The demonstration project took the pond-wetland combination purifica-



Fig. 5 – Guanjinggang vs. Shijiuyang ecological wetland. Upper: Shijiuyang wetland Google Earth image (February 7, 2016), lower: Guanjinggang wetland Google Earth image (February 7, 2016). Yellow line: water boundary; magenta line: drinking water plant; cyan line: sluice gate in the water course or isolating wetland area from the surrounding waters; red line: pipe jacking project underneath the stream; green line and arrow: the main water flow direction. For both wetlands, A: Pretreatment area, B: Wetland root channel ecological purification area, C: Deep purification area. For Shijiuyang wetland, ① Xinchengtang Stream, ② Beijiaohe Stream, ③ Pump station, ④ Water diversion area, ⑤ Shijiuyang water plant, ⑥ Sluice gate, ⑦ Pipe jacking project through Beijiaohe Stream. For Guanjinggang wetland, ① Nanjiaohe Stream, ② Haiyantang Stream, ③ Pump station, ④ Water diversion area, ⑤ Guanjinggang water plant, ⑥ Sluice gate, ⑦ Pipe jacking project through Haiyantang Stream.

tion technology as the foundation (Wang et al., 2016b), and based on the research results of the national water special project under the “Eleventh Five Year Plan” (Wang et al., 2012c, 2013b, 2015a), the technical scheme for upgrading and improving the Guanjinggang ecological wetland was put forward. (1) The pump station was translocated from the middle part of the system (Shijiuyang) to the front end (Guanjinggang) to enhance the efficiency of water cascade and natural aeration and hydraulic regulation (Wang et al., 2015a). (2) Connecting river section, i.e., a meandering channel, was increased after the exit from the wetland root channel ecological purification area to transfer, stabilize and buffer the water quality of previous

area. (3) The simple morphology transformation of the plant-bed/ditch system was the critical point of the whole upgrading (Wang et al., 2015a; Zhang et al., 2017; Su et al., 2018). The plant bed in the wetland root channel ecological purification area was firstly straight deformation and curved. The boundary length of the curved plant-bed/ditch ecotone is about 1.57 times larger than that of the straight one. In order to further increase the effective flow contact area and water filtration efficiency of wetland plant bed, the plant bed was improved from wide bed (generally 18 m in width for a single strip of bed) to narrow bed (generally 7.25 m in width for a single strip). A new small and shallow ditch with a surface width



Fig. 6 – Sampling sites in Guanjinggang ecological wetland during 2014, 2016–2017. Left: Sites in 2014, base map Google Earth image (March 30, 2014); Right: Sites in 2016–2017, base map Google Earth image (February 7, 2016). In 2014, “a”: source water at the entrance of pretreatment area, “h”: exit of wetland root channel ecological purification area; “m”: middle part of deep purification area. In 2016–2017, “A”: source water at the entrance of pretreatment area, “B”: middle part of pretreatment area, “C”: exit of pretreatment area and after overflow weir, “D”: exit of the first patch of wetland root channel ecological purification area, “E”: exit of the second patch of wetland root channel ecological purification area, “F”: exit of the third patch of wetland root channel ecological purification area, “G”: middle part of the fifth patch of wetland root channel ecological purification area, “H”: exit of the whole wetland root channel ecological purification area, “I”: entrance to connecting river section and before submerged spur dikes, “J”: middle part of connecting river section and after submerged spur dikes, “K”: exit of connecting river section, “L”: entrance to deep purification area, “M”: middle part of deep purification area, “N”: exit of deep purification area.

of 3.5 m, a bottom width of 0.5 m and a depth of about 1 m was dug in the center of the established plant bed (Fig. 6, left: wide bed, right: narrow bed). The position of some built gravel or earth filling plugs (or called “gate”), a flow baffle hydraulic structure at the alternate ends of small ditches, was correspondingly adjusted to change the hydraulic flow direction. In the narrow type of plant-bed/ditch, the boundary length of land/water crisscrossed zone was increased by about 1.8 times as much as that of the original wide type, and the area of nitrogen reactive active hot zone in the riparian zone was increased by about 2.5 times. (4) The water checking device in the large ditch (or called “semi-gate”) in the wetland root channel ecological purification area was changed from the stoplog gate to the inverted sluice, in order to control the water distribution ratio more effectively in the plant-bed/ditch system and promote more water flow through the root channel macropores within plant bed (Wang et al., 2015a). (5) Five rows of altogether 303 cages of gabion dams were set up at the overflow weir of the water level rising and falling aeration areas (Zhang et al., 2017). The inside of the dam row by row was filled with gravel, zeolite, calcite 1, calcite 2 and volcanic rock respectively, arranged in a plum blossom pile style, which can enhance the natural oxygen increase of falling wa-

ter current surging, (micro-) biological oxidation of water flow contact and strengthen the adsorption of nitrogen and phosphorus and so on. Three groups of submerged spur dikes were set up in the connecting river section after the water was discharged from the wetland root channel ecological purification area (Zhang et al., 2017). Each group was composed of three submerged spur dikes with a length of 15 m and a width of 4 m. They were arranged in a zigzag pattern, and were filled with calcite and gravel to fine tune and control the water flow in the connecting river section, which can also strengthen the biological oxidation of water flow contact and strengthen the adsorption of nitrogen and phosphorus and so on. The gravel bed and calcite bed platform with a top width of 19 m and a total vertical thickness of 80 cm were added in the coastal zone around the deep purification area to strengthen the contact oxidation and adsorption function of water flow with the shore media (Wang et al., 2015a). (6) In the pretreatment area, large ditches in the wetland root channel ecological purification area, as well as the deep purification area, a total of 622 plots of enclosed floating islands were added, with an area of about 13,302 m². The introduction of winter evergreen plant (*Myriophyllum aquaticum*) can enhance the absorption of nitrogen and phosphorus by aquatic plants during low tempera-

ture period (Liu et al., 2016; Luo et al., 2017). However, attention should be paid to the control of excessive spread and regular vegetation harvesting management of *Myriophyllum aquaticum* (Chen, 2012; Kumwimba et al., 2020).

In the above technical schemes, (1), (2), (4) were finished during the initial construction in 2013, and (3), (5), (6) were finished during the upgrading construction in 2015.

4. Water treatment effects of Guanjinggang ecological wetland

In order to examine the water treatment effect before (2014) and after (2016–2017) the improvement of the Guanjinggang ecological wetland, a third-party monitoring agency was entrusted to independently monitor the generally concerned water quality at the key nodes (mainly confluence area) of the project (Fig. 6). The main physicochemical indexes, determination methods and referred standards were listed in Appendix A Table S1.

4.1. Spatial change of water quality indexes

In 2014 (Appendix A Fig. S1), in the source water, wetland root channel ecological purification area and deep purification area, the basically equal indexes in the three regions were mainly air temperature, water temperature and chloride. The pH value and dissolved oxygen were gradually increased. The indexes with gradient decline included total alkalinity, suspended substance, turbidity, ammonia nitrogen, nitrite nitrogen, nitrate nitrogen, total nitrogen, dissolved reactive phosphorus, total phosphorus, permanganate index, chemical oxygen demand, biochemical oxygen demand after 5 days (BOD₅), iron, manganese, and fecal coliform.

In 2016–2017 (Appendix A Fig. S2), total alkalinity, suspended substance and total nitrogen were the main indexes which were basically equal in the three regions. The gradually increased indexes include pH, dissolved oxygen, chloride, sulfate, calcium hardness, total hardness and nitrate nitrogen. The indexes with gradient decline included air temperature, water temperature, turbidity, ammonia nitrogen, nitrite nitrogen, dissolved reactive phosphorus, total phosphorus, permanganate index, chemical oxygen demand, biochemical oxygen demand after 5 days, iron, manganese, fecal coliform, mineralized degree, total organic carbon and chlorophyll. It can be seen that most of the indexes showed similar change trend in the two operation periods, but a few indexes showed opposite change trend, which reflected the interannual change of wetland purification and the effect of improving measures.

4.2. Seasonal change of water quality indexes

The main indexes of source water and wetland showed obvious and considerably complex seasonal changes (Appendix A Fig. S3, Fig. S4). Here, we take the summer with the highest temperature and the winter with the lowest temperature as examples. In 2014, the indexes of source water and wetland which were high in summer and low in winter mainly included suspended substance, turbidity, dissolved reactive

phosphorus, total phosphorus, chemical oxygen demand, biochemical oxygen demand after 5 days and iron; the indexes of source water and wetland which were low in summer and high in winter mainly included pH, dissolved oxygen, total alkalinity, chloride, ammonia nitrogen, nitrate nitrogen, total nitrogen, manganese and fecal coliform; on the whole, there is little difference for nitrite nitrogen and permanganate index between summer and winter (Appendix A Fig. S3). In 2016–2017, the indexes of source water and wetland which were high in summer and low in winter mainly included turbidity, nitrite nitrogen, dissolved reactive phosphorus, total phosphorus, permanganate index, chemical oxygen demand, fecal coliform and mineralized degree; the indexes of source water and wetland which were low in summer and high in winter mainly included pH, dissolved oxygen, total alkalinity, chloride, sulfate, calcium hardness, total hardness, suspended substance, ammonia nitrogen, nitrate nitrogen, biochemical oxygen demand after 5 days, iron, manganese, chlorophyll; in general, total organic carbon had little difference between summer and winter (Appendix A Fig. S4). It can be seen that some indexes such as suspended substance, biochemical oxygen demand after 5 days and iron showed opposite seasonal trends in the two operation periods, while nitrite nitrogen and permanganate index showed little difference in winter and summer of 2014, and showed high in summer and low in winter in 2016–2017 (Appendix A Fig. S3, Fig. S4). It is worth noting that there were many indexes in Jiaxing plain river network area, which were characterized by low concentrations in summer and high concentrations in winter, while just the opposite the efficiency of wetland treatment generally decreased in winter, which is what we should focus on and overcome.

4.3. Spatiotemporal change differentiation

Furthermore, correlation analysis, analysis of variance and regression analysis (see Appendix A Supplementary Materials and Methods) were used to reveal the seasonal and spatial differences of the main physicochemical indexes and their spatiotemporal variation characteristics. In order to further analyze the correlation between the main physicochemical indexes and temperature, the normality test of the indexes was firstly carried out to decide which correlation analysis was applicable. The Shapiro-Wilk normality test results (Appendix A Table S2 (1), Table S3 (1)) showed that compared with source water, more indexes in the wetland did not follow the normal distribution, which indicated that after the purification of the wetland, the sorting effect of the main indexes in the wetland occurred, and the distribution presented the skewed distribution characteristics; compared with 2014, more indexes in 2016–2017 did not follow the normal distribution, which indicated that after upgrading and improvement of the wetland, the sorting effect and skewed distribution characteristics of main indexes were more apparent. The results of non-parametric Kendall's Tau Correlation (Appendix A Table S2 (2), Table S3 (2)) showed that spatially there were more indexes significantly correlated with water temperature in the wetland compared with source water, and temporally there were more indexes significantly correlated with water temperature in 2016–2017 compared with 2014. Kruskal-Wallis Test results (Appendix A Table S4) showed that in 2014, the seasonal differ-

ences of most indexes reached a significant level among Seasons, while only a few indexes reached a significant level spatially among Stations; in 2016–2017, the seasonal differences of all indexes reached a significant level among Seasons, and half of the indexes reached a significant level spatially among Stations.

In order to semi-quantitatively reveal the changes of main physicochemical indexes with water temperature (indirectly time dimension, seasonally) and the length of wetland hydraulic flow pathway (space dimension, horizontally) as well as the relative contribution of these two variables, a stepwise regression analysis was performed. The results of stepwise regression analysis (Appendix A Table S5) showed that most of the physicochemical indexes were determined more or less by water temperature during the two operation periods, and the contribution rate (coefficient of determination) ranged from 2.5% to 73.2%, with an average contribution rate of 31.3%, i.e., about one-third. The changes of a considerable number of indexes were determined by the length of wetland hydraulic flow pathway, and its contribution rate ranged from 0.86% to 44.2%, with an average contribution rate of 9.6%, i.e., about ten percent. The specific results were as follows: in 2014, ① the indexes determined by water temperature exceeding the length of wetland hydraulic flow pathway included suspended substance, ammonia nitrogen, dissolved reactive phosphorus, chemical oxygen demand and manganese; ② the indexes determined by the length of wetland hydraulic flow pathway exceeding water temperature were total phosphorus, iron and fecal coliform; ③ the indexes only significantly determined by water temperature included pH, dissolved oxygen, chloride, nitrate nitrogen, total nitrogen, and biochemical oxygen demand after 5 days; ④ turbidity was only significantly determined by the wetland hydraulic flow pathway length; ⑤ total alkalinity, nitrite nitrogen, and permanganate index were not significantly determined by water temperature or wetland hydraulic flow pathway length. In 2016–2017, ① the indexes that were determined by water temperature exceeding the length of wetland hydraulic flow pathway included dissolved oxygen, total alkalinity, suspended substance, turbidity, ammonia nitrogen, nitrite nitrogen, nitrate nitrogen, total nitrogen, dissolved reactive phosphorus, total phosphorus, permanganate index, chemical oxygen demand, biochemical oxygen demand after 5 days and iron; ② the indexes determined by the length of wetland hydraulic flow pathway exceeding water temperature were manganese and fecal coliform; ③ the indexes only significantly determined by water temperature included pH, chloride, sulfate, calcium hardness, total hardness and mineralized degree; ④ chlorophyll was only significantly determined by the wetland hydraulic flow pathway length; ⑤ total organic carbon was not significantly determined by water temperature or wetland hydraulic flow pathway length.

4.4. Process section reduction contribution

In terms of the reduction contribution rate of each process section, in 2014, the reduction of turbidity, ammonia nitrogen, nitrite nitrogen, nitrate nitrogen, total nitrogen, dissolved reactive phosphorus, total phosphorus, permanganate index, iron, and fecal coliform as well as the improvement of dissolved oxygen, mainly occurred in the wetland root channel

ecological purification area (including pretreatment area); the reduction of suspended substance and biochemical oxygen demand after 5 days mainly occurred in the deep purification area; the reduction contribution rates of chemical oxygen demand and manganese in the wetland root channel ecological purification area and deep purification area were roughly equal (Appendix A Table S6). Water pollution comprehensive index (WQII, I_p) is a comprehensive index reflecting the overall water pollution status, which has been widely used in comprehensive evaluation of water quality. Its calculation method was shown in the Supplementary Materials and Methods. In 2014, WQII of source water, wetland root channel ecological purification area and deep purification area were 4.56 ± 0.08 , 3.59 ± 0.15 and 3.50 ± 0.12 (mean $\pm$ S.D.), respectively. The ecological wetland promoted the water quality of source water from class IV to class III, and the main improvement process occurred in the wetland root channel ecological purification area (91.5%). The comprehensive contribution of water quality improvement in the deep purification area was about 8.5%. In 2016–2017, the reduction of ammonia nitrogen, nitrite nitrogen, dissolved reactive phosphorus, total phosphorus, permanganate index, chemical oxygen demand, manganese, and fecal coliform as well as the improvement of dissolved oxygen, mainly occurred in the wetland root channel ecological purification area (including pretreatment area); the reduction of suspended substance, turbidity, biochemical oxygen demand after 5 days, iron, mineralized degree, total organic carbon, and chlorophyll mainly occurred in the deep purification area; while the wetland did not remove nitrate nitrogen and total nitrogen by annual average (Appendix A Table S7). In 2016–2017, WQII of source water, wetland root channel ecological purification area and deep purification area were 4.31 ± 0.07 , 3.32 ± 0.09 and 3.29 ± 0.06 (mean $\pm$ S.D.) respectively. The ecological wetland promoted the water quality of source water from class IV to class III, the main improvement process occurred in the wetland root channel ecological purification area (97.1%), and the comprehensive contribution of water quality improvement in the deep purification area was about 2.9%.

4.5. Seasonal reduction contribution

The seasonal reduction and contribution rate of each process section of the wetland were further explored. The seasonal changes of indexes and contributions were shown in Appendix A Tables S8 and S9 in detail, which would not be described extensively here. We could see the comprehensive results nonetheless the removal rate of some indexes was not ideal in some seasons. In 2014, the order of absolute reduction of water pollution comprehensive index in four seasons was: autumn > spring > winter > summer. The contribution rates of wetland root channel ecological purification area (including pretreatment area) and deep purification area in four seasons were: spring (91.2% vs. 8.8%), summer (86.9% vs. 13.1%), autumn (91.2% vs. 8.8%) and winter (96.4% vs. 3.6%) (Appendix A Table S8). In 2016–2017, the order of absolute reduction of water pollution comprehensive index in four seasons was: spring > winter > autumn > summer. The contribution rates of wetland root channel ecological purification area (including pretreatment area) and deep purification area in four seasons were: spring (100.0% vs. 0.0%), summer (94.8% vs. 5.2%), au-

turn (100.0% vs. 0.0%) and winter (99.0% vs. 1.0%) (Appendix A Table S9). Compared with 2014, in 2016–2017, the contribution proportion of wetland root channel ecological purification area was further increased from 91.5% to 97.1% after the improvement of relevant engineering measures. Interestingly, in the relatively low temperature seasons including spring, autumn, winter, the wetland could still have a relatively better removal rate, however in the warmer summer, the absolute reduction amount of water pollution comprehensive index was the lowest. At the same time, we could also see that the reduction of water pollution comprehensive index in ecological wetland during four seasons was quite robust and stable. Although the contribution rate of wetland root channel ecological purification area (including pretreatment area) during the four seasons was absolutely dominant, from the specific indexes, the deep purification area could still play a complementary role. In addition, it also reflected the function of stabilizing the water quality of the deep purification area, so as to store the purified clean water and supply water for the waterworks.

Generally speaking, the above results showed that the seasonal variation of source water in plain river network area was large, the water quality was temporally unstable, and the water quality indexes had some possibility of abrupt changes in pulse due to the upstream “guest” water and local pollution, so it was difficult for drinking water plants to timely buffer and respond to the changes. This is why ecological wetlands should be built before water plants for buffer and mitigation, but how to improve the stability and strong buffer of wetlands has become an important challenge for our scientific and technological personnel.

4.6. Comprehensive evaluation function value

In order to comprehensively reflect the overall change trend of main physicochemical indexes in the two operation periods, or to reflect the comprehensive purification capability of wetlands, we use the principal component to establish a comprehensive evaluation function. It is generally believed that this method is superior to any other comprehensive evaluation methods at present. The calculation method and process were described in detail in the supplementary information (Appendix A Table S10). The comprehensive evaluation function value (F) (Fig. 7) showed that in 2014 and 2016–2017, the water quality of wetland source water was quite different, and the water quality of wetland source water was worse in 2014. In 2016–2017, under the action of water control strategies such as “five water co-governance” and “river chief system” in Jiaying, the water quality of wetland source water had been significantly improved, and the absolute value of F value had decreased by 0.55 units (Fig. 7). The common corresponding source water (Sites “a”, “A”), wetland root channel ecological purification area (Sites “h”, “H”) and deep purification area (Sites “m”, “M”) during two operation periods (Stages “Before”, “After”) were compared. It was showed that in 2014, the F value of source water (Site “a”), wetland root channel ecological purification area (Site “h”) and deep purification area (Site “m”) gradually decreased from 0.71 to 0.34 and then to 0.12 (Fig. 7). In 2016–2017, the F value still showed a gradual decrease trend along the wetland hydraulic flow pathway; the

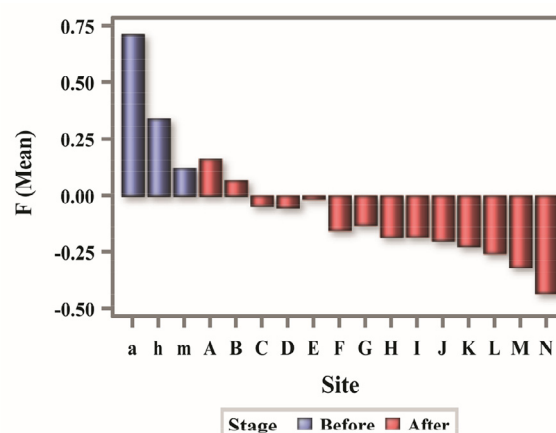


Fig. 7 – Comprehensive evaluation function value (F) based on the principal component analysis of the major water quality indexes in Guanjiangang ecological wetland during 2014, 2016–2017. Stage “Before”: Sites “a”, “h”, “m” in 2014 before upgrading and improving the Guanjiangang ecological wetland; Stage “After”: Sites “A” – “N” in 2016–2017 after upgrading and improving the Guanjiangang ecological wetland.

F value of source water (Site “A”), wetland root channel ecological purification area (Site “H”) and deep purification area (Site “M”) decreased from 0.16 to –0.18 and then to –0.32 (Fig. 7).

The results of SAS random forest analysis (PROC HPFOREST) (Cary, NC: SAS Institute Inc.) showed that, among the indexes for the establishment of comprehensive evaluation function, the first four variables with the highest importance to the value of comprehensive evaluation function were chemical oxygen demand, manganese, permanganate index and ammonia nitrogen (in order of importance) (Appendix A Table S11), which roughly reflects the basic characteristics of organic micropollution and ammonia nitrogen pollution in Jiaying river network area.

4.7. Ammonia nitrogen removal effect

The water purification effect was integrally analyzed focusing on ammonia nitrogen ($\text{NH}_4^+ \cdot \text{N}$) removal in Shijiuyang ecological wetland and Guanjiangang ecological wetland as well as the latter before and after the improvement.

4.7.1. 2014 Guanjiangang, year-round vs. winter

According to the results of the first period annual monitoring in 2014 (Appendix A Fig. S1), in Guanjiangang ecological wetland, the water quality purification was mainly completed by the wetland root channel ecological purification area, where the ammonia nitrogen was reduced from 1.26 ± 0.97 mg/L to 0.76 ± 0.78 mg/L (mean $\pm$ S.D.), and the removal rate was averagely 39.7%. The ammonia nitrogen was reduced from 0.76 ± 0.78 mg/L to 0.68 ± 0.65 mg/L (mean $\pm$ S.D.) through the connecting river section after the wetland root channel ecological purification area as well as the deep purification area, and the re-removal rate was 10.5%. The annual average removal rate of ammonia nitrogen was 46.0%. The average

removal rate of ammonia nitrogen in the winter low temperature season of 2014 was 45%, ranging from 21.6% to 78.4%, and the average removal rate of wetland root channel ecological purification area was 34%, accounting for the main force purification.

4.7.2. 2014 Guanjinggang vs. Shijiuyang, same period and duration, year-round and winter

Both wetlands were compared. Firstly compared with the same period, in 2014, the annual average value of ammonia nitrogen in the influent and effluent of Shijiuyang ecological wetland was 1.15 mg/L and 0.78 mg/L respectively, with an average removal rate of 32.2%. The average removal rate of Guanjinggang ecological wetland in 2014 was 46.0%, which was higher than 32.2%, about 14 percentage points. Compared with the same operation duration, the Guanjinggang ecological wetland was put into trial operation in July 2013, and Shijiuyang ecological wetland was put into trial operation in July 2008. In the same operation duration, the year round operation data of 2009 in Shijiuyang ecological wetland were used to compare the removal. In 2009, the average value of ammonia nitrogen in the influent and effluent of Shijiuyang ecological wetland was 1.29 mg/L and 0.86 mg/L respectively, with an average removal rate of 33.3%. The average removal rate of Guanjinggang ecological wetland in 2014 was 46.0%, which was higher than 33.3%, about 13 percentage points.

Then we compared the winter removal rate of ammonia nitrogen in Guanjinggang and Shijiuyang ecological wetlands. Compared with the same period, in winter 2014, the average value of ammonia nitrogen in the influent and effluent of Shijiuyang ecological wetland was 1.69 mg/L and 1.79 mg/L, respectively. In terms of the average value of ammonia nitrogen, the wetland had no positive removal rate. The average removal rate of ammonia nitrogen was 45% in winter 2014 in Guanjinggang ecological wetland. Compared with the same operation duration, in winter 2009, the average value of ammonia nitrogen in the influent and effluent of Shijiuyang ecological wetland was 1.90 mg/L and 1.62 mg/L respectively, with an average removal rate of 14.7%. In winter 2014, the average removal rate of ammonia nitrogen in Guanjinggang ecological wetland was 45% (range: 21.6%–78.4%), which was higher than 14.7%, about 30 percentage points.

4.7.3. 2016 vs. 2014 Guanjinggang, initial improvement and additional upgrading

In order to further examine the purification effect of each functional area and structural unit after the implementation of the additional upgrading technology in Guanjinggang ecological wetland, a new round of one-year periodic monitoring had been carried out between April 2016 and March 2017, and the monitoring points were increased (Fig. 6), and the water chemistry indexes were expanded (Appendix A Fig. S2). The results were shown in Appendix A Fig. S2. Here in order to avoid the interannual difference caused by cross year, we used the data within the same natural year for calculation and comparison. From April to December 2016, the average removal rate of ammonia nitrogen was 55.8%, ranging from 24.9% to 90.2%; in winter (December 2016), the removal rate of ammonia nitrogen was 29.3%. These removal rates were higher than Shijiuyang ecological wetland in the same period and

operation duration. The monitoring results showed that the removal of ammonia nitrogen was mainly completed in the front of the wetland, accounting for 86.4% of the ammonia nitrogen removal contribution rate on average, and the wetland root channel ecological purification area had the function of mitigating and buffering water quality. This showed that the three-dimensional gabion reinforcement of overflow weir, arrangement of floating plants and morphology transformation of plant bed at the front end of the wetland had played an effective role, and the effect was obvious compared with 2014.

Then, the Guanjinggang ecological wetland was compared in the same month span before and after the wetland improvement from April to December 2014 and April to December 2016. From April to December 2014, the average concentrations of ammonia nitrogen in the source water (Site “a”), the effluent from the wetland root channel ecological purification area (Site “h”) and the water in the deep purification area (Site “m”) were 0.800 ± 0.571 , 0.347 ± 0.182 and 0.334 ± 0.207 mg/L (mean $\pm$ S.D.), respectively. The absolute reductions of ammonia nitrogen concentration in wetland root channel ecological purification area and deep purification area were 0.453 mg/L and 0.013 mg/L, respectively. The total removal rate of ammonia nitrogen, the removal rates of wetland root channel ecological purification area and deep purification area were 58.2%, 56.6% and 1.6% respectively, and the removal rate of wetland root channel ecological purification area accounted for 97.2% of the total removal rate. From April to December 2016, the average concentrations of ammonia nitrogen in the source water (Site “A”), the effluent from the wetland root channel ecological purification area (Site “H”) and the water in the deep purification area (Site “M”) were 1.061 ± 0.701 , 0.536 ± 0.561 and 0.484 ± 0.552 mg/L (mean $\pm$ S.D.), respectively. The absolute reductions of ammonia nitrogen concentration in wetland root channel ecological purification area and deep purification area were 0.525 mg/L and 0.052 mg/L, respectively. The total removal rate of ammonia nitrogen, the removal rates of wetland root channel ecological purification area and deep purification area were 54.4%, 49.5% and 4.9% respectively, and the removal rate of wetland root channel ecological purification area accounted for 91.0% of the total removal rate. It should be noted that the concentration of ammonia nitrogen in 2016 (1.061 mg/L) was 32.6% higher than that in 2014 (0.800 mg/L), but the total removal rate of ammonia nitrogen only decreased from 58.2% to 54.4% (only reduced by 6.5%), and the absolute removal amount of ammonia nitrogen in 2016 was both significantly higher than that in 2014 in the two functional areas. It is known that the levels of nitrogen concentration can also affect the removal efficiency (Kumar et al., 2020b).

4.7.4. Ammonia nitrogen removal enhancement and comprehensive water treatment effect

The microbial nitrogen process results showed that the substantial ammonium oxidation rate could be obtained by enhancing ammonia oxidizing bacteria activity rather than ammonia oxidizing archaea activity (Su et al., 2018). In the plant-bed/ditch system, i.e., wetland root channel ecological purification area, ditch center and plant-bed fringe were two active zones for ammonia nitrogen removal with ammonium oxidation rate peaking at 2.98 ± 0.04 mg N/kg/day and 2.15 ± 0.02 mg

N/kg/day, respectively (Su et al., 2018). The enhanced ammonia oxidizing bacteria activity were achieved by increasing water level fluctuations, extending hydraulic retention time, and stimulating substrate availability, which subsequently enhanced ammonia nitrogen removal by 34.1% in Guanjinggang ecological wetland (Su et al., 2018). Although the ammonia oxidizing archaea activity decreased overall with extended hydraulic retention time and increased ammonia nitrogen contents in Guanjinggang ecological wetland, ammonia oxidizing archaea still played a major role on ammonium oxidation in plant-bed soil especially during the cold season (Wang et al., 2013a; Zou et al., 2014; Su et al., 2018). The separate contributions of the updated technologies for water quality improvement in Guanjinggang ecological wetland were examined individually. As for the removed mass of ammonia nitrogen, the contribution of plant-bed/ditch morphology reform (sustainable) accounted for the largest 36%, the winter evergreen plant uptake on the premise of regular harvesting 24%, stone gabion, submerged groin, and riparian stone platform considering the saturation 12%, as well as the sedimentation, adsorption, and other water column process 28%.

Water source ecological wetland has a broad spectrum of water treatment effect. The long-term monitoring results of Guanjinggang ecological wetland by the drinking water plant from July 2013 to November 2019 showed that under the hydraulic loading rate $0.3\text{--}0.4\text{ m}^3/(\text{m}^2\cdot\text{day})$ and hydraulic retention time 2–3 days, the lower quartile, upper quartile (quartile range) and median (in brackets) removal rates for common water quality indexes were as follows, turbidity 11.2%–44.7% (29.3%), ammonia nitrogen 25.0%–75.7% (47.1%), unionized ammonia 23.7%–77.6% (49.6%), nitrite nitrogen 9.4%–59.5% (30.8%), permanganate index 5.2%–15.3% (10.5%), iron 5.2%–38.6% (23.3%), and manganese 5.3%–32.0% (19.4%).

4.8. Cost and multifunctional services

Constructed wetlands have been increasingly regarded as a promising green natural treatment technology with low construction investment, low maintenance, and high efficiency (e.g., Stefanakis, 2016; Carrillo et al., 2019; Nuamah et al., 2020; Khan et al., 2020; Kumar et al., 2020a). The vital environmental role of a wetland in the environment is a natural water filter (Khan et al., 2020). The investment and operation cost of traditional constructed wetland is only 1/10–1/2 of that of traditional secondary wastewater treatment plant (Liu, 2010). It has wide application value, being especially suitable for the suburbs with relatively backward economic development, small and medium-sized towns, and vast rural areas. The specific investment costs are different depending on the geographical location, geological conditions and the wetland substrate used, but generally, the investment cost of surface flow constructed wetland construction is about 150–200 RMB/m², and that of subsurface flow constructed wetland is about 200–300 RMB/m² (Liu, 2010). The engineering construction cost of Guanjinggang ecological wetland, a semi-subsurface flow wetland, was 155 RMB/m² footprint area, and the operation and management costs were about 0.055 RMB/m³ treated water, which were both lower than the costs of traditional constructed wetlands.

The operation of ecological wetlands ameliorated the source water quality, and reduced the drinking water treatment reagents, thereby bringing about economic benefits. At the same time, through wetlands construction and operation, people can see how the micro-polluted surface water becomes clear and clean, so promoting a significant social benefit. In addition, wetlands as a viable component of urban green space, could beautify the regional eco-environment, freshen the air, increase the urban ecological taste, and enhance the eco-environmental protection publicity (Yin et al., 2010). Thus, the multifunctional service values and indirect benefits are substantial.

5. Model of water source ecological wetland with constructed root channel technology (CRCT)

This study systematically reported a kind of water source ecological wetland with the constructed root channel wetland (CRCW) as the core, aiming to provide a typical paradigm for countries or regions with micro-polluted water sources in the world for reference. The conceptual diagram of the water source ecological wetland is shown in Fig. 8, which can be analogized as “human-body wetland model” based on bionics or biomimetics. The main strengthened structural units are also shown. Through optimized design and strict construction as well as proper operation and maintenance, the ecological wetland system is an organic whole just like the human body system. It can operate coordinately, flexibly and cooperatively, and has the function of continuous self-healing and regeneration cycle. The wetland components can coordinate with each other in a systematic way, so that an extensive range of complex biochemical reactions can normally occur in the wetland, and play a very good purification function for various kinds of compound pollutants in the source water.

The whole area of wetland can be generalized into three functional areas (Fig. 8), namely (1) pretreatment area, (2) central treatment area, and (3) post-treatment area. The mechanism of each functional area has been mentioned and discussed in the context. Here we further emphasize the mechanism of action as follows. ① The pretreatment area is the first throat and fortress of wetland. Its well-tuned functions of barrier, interception, natural aeration, preferential settlement and irregular dredging can provide critical premise and guarantee for the stability and long-term operation of central wetland. The pretreatment area is roughly equivalent to the “throat” of human body, the “lung” of respiratory system and the “stomach” of digestive system. In this regard, the ecological filter dam (considering multiple), multistage pond and weir, submerged groin, overflow weir, three-dimensional gabion dam and other hydraulic structures as well as reasonable design of deep pool and shallow shoal play their respective and combined functions actively. The pump station acts as the “heart” of wetland hydrodynamic force, pumping water into the wetland continually and providing the power of water head gradients for wetland operation. ② The central treatment area is the main purification unit of the whole wetland, in which the creative mechanism of constructed root channel wetland is used. The crisscrossed and meandering “multi curve” interfacial structure in this area pro-

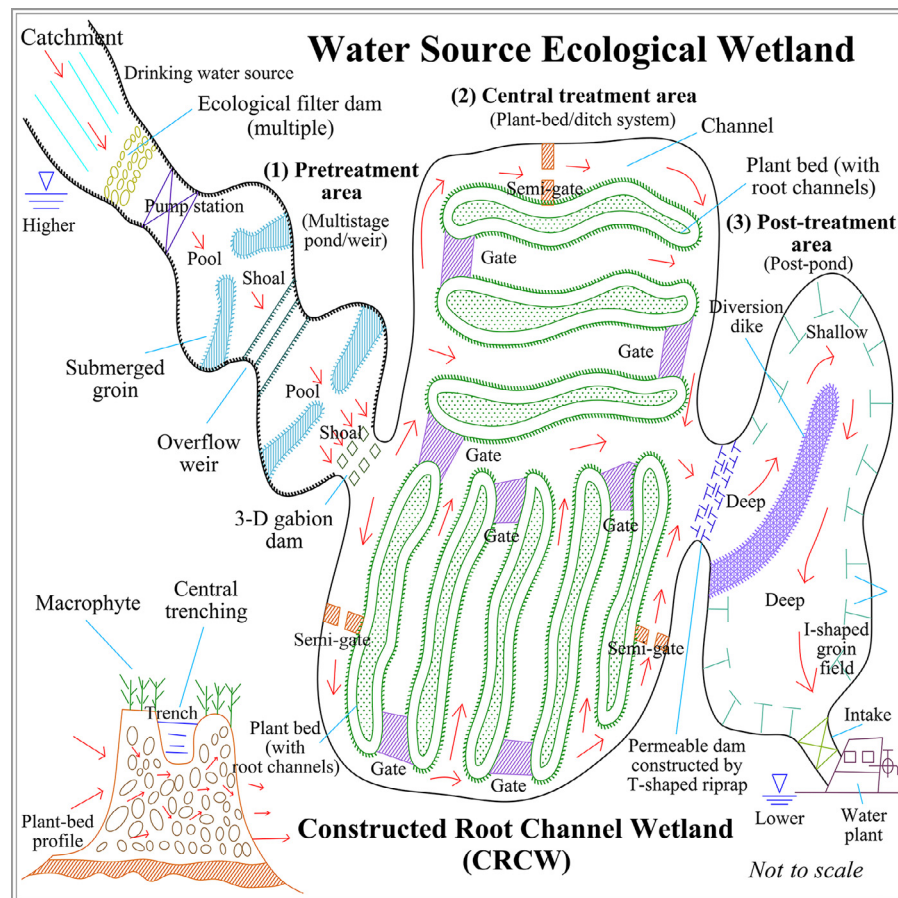


Fig. 8 – Conceptual diagram of water source ecological wetland with constructed root channels.

longs the hydraulic retention time of wetland efficiently, enhances the effective contact between water and soil, and alternates the oxidation and reduction environment continually. There are high-efficiency reaction zones of various substances, reactions, multiphase and polymorphic transformation in this area, where the most complex water purification mechanisms likely occur therein. Here, the hydraulic microstructures such as earth/rock dam (“gate”, either impermeable or semi-permeable) and “semi-gate” (acting as water checking device) play a role in regulating and redistributing the water flow into the wetland root channel ecological purification area, strengthening the percolation process of water flow in the root-channel plant bed. The trenching practice in the center of the plant bed can simply and significantly improve the water flux between the root-channel plant bed and the adjacent ditches, and greatly increase the area and volume of the reactive area of the wetland under the promotion of hydraulic control measures (Fig. 8). The central treatment area is equivalent to the “stomach” of the digestive system and the “kidney” of the urinary system (glomerulus, renal tubules, etc.). ③ The post-treatment area is last but not the least as an indispensable part for the water source ecological wetland, which mainly stores the purified clean water, continually supplies the daily water for the waterworks, and uses its huge storage capacity as the buffer and emergency needs. In this area, permeable dam constructed by the T-shaped riprap, I-

shaped groin field, diversion dike and other hydraulic structures can make this area still have enhanced water purification function. The post-treatment area is roughly equivalent to the “bladder” organ of human urinary system. Wetland system is similar to human body system, but slightly different is that the urinary system of human body is “Accept the clean and Spit out the dirty”, while the wetland system is “Accept the dirty and Spit out the clean”.

6. Core mechanism and extension of water source ecological wetland

In natural wetland, ecological systems are the base for the treatment system in constructed wetland (Khan et al., 2020). Humans designed engineered ecosystem which is termed as constructed wetland to remove contaminants from wastewater treatment that can mimic the habitat and hydraulic condition occurring in the swamp (Khan et al., 2020). Constructed wetlands are natural systems which treat wastewater under different operation conditions taking advantage of the processes that involve different components such as vegetation, soil, and microorganism (Carrillo et al., 2019). The artificially constructed wetland system resembles the treatment that occurs in natural wetland by relying on a combination of naturally occurring physical, chemical, and biological pro-

cess and on the aquatic plants and heterotrophic microorganisms (Olejnik and Wojciechowski, 2012; Khan et al., 2020). The mechanisms of wastewater treatment in wetlands are extremely complicated, including a series of physical, chemical, and biochemical processes (Khan et al., 2020). The wetland efficiency to remove the pollutants from the wastewater primarily depends on the root zone relations between soil, pollutants, plant roots, and diversity of microorganisms (Olejnik and Wojciechowski, 2012). The simple idea behind constructed wetlands is their ability to absorb, filter, and metabolize dissolved and suspended matter (Khan et al., 2020).

This study reports a new type of large scale nature-mimicking wetland system, which is simple and natural, and provides a paradigm for water source ecological wetland restoration. Based on the long-term study on the structure and function of the reed-dominated land/water ecotone, it was found that the boundary (edge, fringe) filtration effect of the land/water ecotone was the result of the coupling of root channel interaction and water level fluctuation (Wang and Yin, 2008). The purification mechanism of root channel macro-pore and multi-media in the soil of the land/water ecotone was clarified, and a method of constructing ecological root channel of wetland was invented, which provides a basis for the large-scale construction of ecological wetland (Wang et al., 2012a). It opens up a new way of design and application of ecological wetland. Further study found that the boundary area between plant bed and ditch is the key active area ("hot area") for pollutant removal, and the synergistic effect of rhizospheric microorganisms and plants can significantly broaden the purification species and low temperature tolerance range of pollutants mitigation (Su et al., 2018; Wang et al., 2018). Therefore, a series of new technologies had been created to enhance the active hot area of wetland ecotone boundary as was stated in this paper, and the active hot area has been stimulated and strengthened by means of water level regulation. This paper innovatively combines the constructed wetland ecological root channel technology (i.e., CRCT), enhanced plant-bed/ditch system (i.e., PBD system) and the greenhouse gas emission mitigation effect of wetland ecological engineering, and puts forward a new idea of micro-polluted water quality control by using the edge filtration effect of land/water ecotone and key living areas for pollutant degradation and transformation (Wang et al., 2016b). Low greenhouse gas emissions compared to other conventional methods while treating the wastewater is an advantage for constructed wetland treatment system, and in particular, hybrid constructed wetland systems attain high removal efficiency with low greenhouse gas emission (Kumar et al., 2020a). Among the constructed water source ecological wetlands, the wetland root channel ecological purification area and the plant-bed/ditch system are the core and main forces of various purification processes, which harbor significant mechanisms of sedimentation, precipitation, flocculation, adsorption, absorption, trapping, enrichment, assimilation, filtration, redox reaction alternation and activation (aerobic, anoxic and / or anaerobic zones), as well as biochemical degradation and so on (Zheng et al., 2012; Wang et al., 2012b, 2012c, 2013a, 2013b, 2014a, 2014b, 2015a, 2016b; Su et al., 2018; Wang et al., 2018, 2019, 2020). Moreover, they are also important reactor of the wetland organic matter biodegradation (Wang et al., 2013b) as well as

the algae traps and degradation (Zheng et al., 2012), in which the optimized hydraulic distribution and regulation mechanism are particularly critical (Wei et al., 2018). A key point that is often overlooked is the supply of organic matter by plants (Khan et al., 2020). Organic matter derived from plants in wetlands constantly provides carbon sources for bacterial metabolism as well as metal sorption thus encouraging long-term operation (Marchand et al., 2010; Khan et al., 2020). With wetland succession, retention times become high with increasing vegetation resulting in increased sedimentation, but it may absorb metal and other types of suspended material (Khan et al., 2020). With these complex mechanisms, it can be understood that generally the removal efficiency of wetland in winter was considerable or even better than that in summer. Other researcher also pointed out that the reactions for phosphorus sorption have lower temperature effect, however, the removal of nitrogen, phosphorus and particulate carbon by physical processes are not much affected by temperature (Kumar et al., 2020b). The complex, multi-phase and multi-media action mechanisms of wetland can provide guarantee and strong buffer for it to deal with the combined pollution of source water and emerging pollutants released to the environment.

Based on the construction of the first large-scale ecological water source wetland (Shijiuyang ecological wetland, 110 hm²) in China in 2008, five large-scale water source ecological wetland projects have been built in the water polluted river network areas in Jiaxing, Haining, and Tongxiang in recent 10 years, forming a drinking water source wetland group in Northern Zhejiang, which has a significant radiation effect of "technology leading project". The constructed water source ecological wetlands can upgrade the comprehensive class of source water quality by more than one level, reduce the risk of combined pollution of multiple pollutants coexistence by about 40%–60%, effectively improve the quality of drinking water source, greatly enhance the capability of water source security, and show multiple ecological service functions. Relevant research and engineering application results have created an advanced mode of ecological governance and restoration of urban and rural drinking water sources in China, led the transformation of China's drinking water source construction from traditional engineering to ecological restoration, and provided important technical basis and demonstration cases for developing countries to solve water quality problems.

According to the previous research results, the water quality purification process of water source ecological wetland is variable, complex and multi-dimensional. The purification process and effect of wetland on the main indexes of water source are mainly affected by the seasonal changes within the year, the interannual factor changes, the impact of pollutant concentration and pulse load in water source, the ecological succession process of wetland itself and the water supply of wetland to waterworks. In order to deeply and systematically reflect the purification process and mechanism of wetland system, we need to use interdisciplinary technical means, and carry out a comprehensive and systematic design in the space-time dimension, so as to better reveal the water purification and retention process mechanism of wetland water phase, sediment phase and biological phase. The sus-

tained in-depth studies and findings will present a reference for the development of ecological water source and provide a strong guidance for water quality improvement technology and long-term operation mechanism.

7. Conclusions

The source water of Jiaying in the dense plain river network of eastern China shows typical organic micro-pollution as well as complex, changeable and unstable characteristics due to the multiple sources of pollution, variable hydrodynamic and catchment terminal. This study indicated that the ecological wetland paradigm could effectively alleviate water pollution, and cope with the single water source, unsteadiness, and pulse load in the river network areas. Pond-wetland complex, plant-bed/ditch system and root channel multistage interception are the technical foundation of water quality improvement scheme for the plain river network area. Shijiyang ecological wetland has created the first water source ecological wetland in China. On this basis, the further improved Guanjinggang ecological wetland had a larger scale and better water quality treatment effect, and the ammonia nitrogen removal rate in the wetland root channel ecological purification area had increased significantly. Jiaying water source ecological wetland also has multiple ecological service functions, such as promoting biodiversity, improving local microclimate, upgrading the quality of ecological environment, and serving as the ecological and environmental protection education base. At present, the water source ecological wetlands which have been put into operation in Jiaying, Haining, and Tongxiang cover an area of about 10 km², and the total daily water supply quantity are about 1,750,000–1,900,000 m³. More people benefit from them.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Supplementary data

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jes.2021.03.015.

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